

Executive Summary

Dmitrii Kuptsov

Insaf Ibragimov

Sofia Pirogova

June 18, 2026

Objective. We classify English hotel reviews as fake (a paid crowdworker who never stayed) or genuine (scraped from TripAdvisor), using Ott et al., 1600 balanced reviews, a standard benchmark since 2011. We lead with macro-F1. The literature headline is the 30 point gap between models near 90 to 91 percent and human judges at 53 to 62 percent.

Baselines (Part 1). A random classifier scores 0.493 macro-F1, the 50 percent floor. Our rule-based model reaches 0.753 from four data-derived features, two log-odds lexicons plus noun and pronoun rates, and passes the human range. Truthful reviews name concrete things and use more nouns. Deceptive reviews use more pronouns and promotional words.

32-label regime (Part 2). Plain BERT on 32 examples reaches 0.677, above random and below the rule-based baseline. EDA augmentation raises BERT to 0.779, our best low-resource result. The zero-shot LLM (qwen3-32b) scores 0.533 and few-shot 0.597, barely above random. Deceptive-only LLM generation reaches 0.744. The rule model refit on 32 examples drops to 0.538. EDA plus LLM generation (0.685) does worse than EDA alone, so BERT plus EDA is our optimal low-resource technique.

Full data (Part 3). BERT rises from 0.513 at 1 percent to about 0.88 at 75 percent, then levels off, with most of the gain in the first few hundred examples. The best single point is BERT with EDA plus LLM generation at 0.903 (75 percent of the data), within one point of the BERT SOTA near 0.91 and above the feature SVM near 0.90. No model reaches the 0.91 SOTA.

Compression (Part 4). We compress the full-data BERT teacher (0.859 macro-F1, 40.6 samples per second, 438 MB). Distillation into DistilBERT runs at 87.5 samples per second, a 2.2x speedup, and 268 MB, a 1.6x reduction, while macro-F1 holds at 0.852. Both targets are met. Dynamic int8 quantization goes the wrong direction on speed at 10.5 samples per second, far below the teacher on CPU. DistilBERT adds 12 new errors against the teacher, in both directions, flagging some genuine reviews as fake and passing some fakes as genuine.

Takeaway. We move from 0.49 with random guessing to 0.90 with the best augmented BERT, a gain of 0.41 F1 points. Deceptive opinion spam is detectable, but requires strong pretrained representations or substantial labeled data, not simple surface rules alone.